OpenCourant examples in FreeCAD: a rough analysis of results and meshes

Generated by an AI assistant (Claude, Anthropic) for the FC_OpenCourant project

1 November 2026

AI-generated – not verified by a human

This document was written by an AI model from automated test runs of the FC_OpenCourant workbench (OpenCourant / OpenRadioss in FreeCAD). The numbers were read from solver listings and result files by scripts, and the interpretations and conclusions are the model's own. **No human engineer has checked them.** Do not use them for design, certification or any decision with safety or financial consequences. The document comes with **no warranty** (Section 8).

1 Scope and set-up

Three example models shipped in `examples/` were solved and the results compared with simple reference solutions; the meshes are assessed roughly. A side study compares tetrahedral meshes with CalculiX.

- Software: FreeCAD 1.0.0 (Debian) and 1.1.4 (Flatpak), workbench version 0.0.1, OpenCourant `latest-20261011` (starter commit `cb760c206e`), Debian 13, x86-64.
- Units throughout: mm, t, s, N, MPa. Steel: $E = 210\,000$ MPa, $\nu = 0.3$, $\rho = 7.9 \cdot 10^{-9}$ t/mm^3 (Taylor bar: $7.85 \cdot 10^{-9}$).
- Reproduction: the models are rebuilt by `examples/build_examples.py` (run with `freecadcmd`), run in the FreeCAD GUI by `tests/run_gui_test.sh`, and the unit and solver tests by `tests/tests.py`.
- Each case was run once; there is no systematic mesh convergence study.

2 Cantilever beam (FEM workbench, hexahedra)

Model and mesh. Beam $100 \times 10 \times 10$ mm, clamped at $x = 0$, $F = 100$ N downwards at the free end, ramped linearly over 10 ms. Structured mesh of $20 \times 2 \times 2 = 40$ eight-node hexahedra (189 nodes), cubes of 5 mm (aspect ratio 1), `ELFORM 1` (one integration point, physical hourglass control). No mass scaling.

Numerics. Time step 0.716 μs (estimate from the dilatational wave speed ≈ 6.0 km/s and 0.9×5 mm: 0.75 μs), 13 977 cycles, 1.6 s wall time. Mass $7.90 \cdot 10^{-5}$ t as expected. At the end the kinetic energy is 0.05 % of the internal energy (9.03 N mm), so the response is quasi-static; the external work (9.04 N mm) matches $F\delta/2 = 9.3$ N mm within a few per cent.

Results. Tip deflection 0.1868 mm against 0.192 mm from beam theory with shear deformation (−2.7 %). The largest nodal von Mises stress is 26.9 MPa, while the bending stress at the clamped surface is $\sigma = Fl\,c/I = 60$ MPa. This is expected for this mesh: with one integration point per element the stress is sampled at the element centre, 2.5 mm from the neutral axis and 2.5 mm from the clamp, where beam theory gives 29 MPa; averaging to the nodes lowers it further.

Mesh assessment. Adequate for the deflection; too coarse for stresses. For surface stresses use at least 4–8 elements through the thickness or fully integrated elements, and do not read peak stresses from one-point elements next to a clamp.

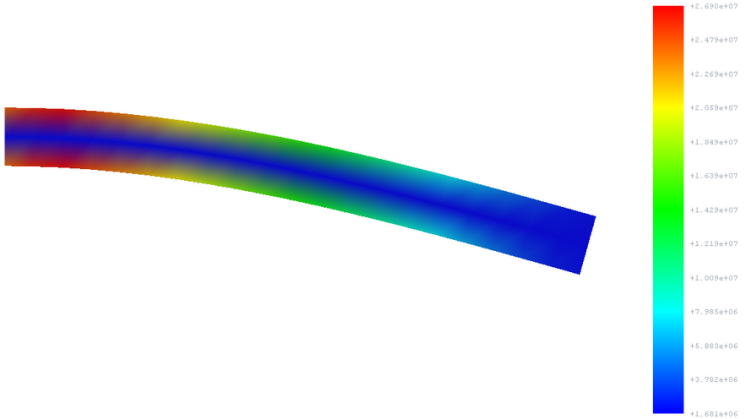


Figure 1: Cantilever, von Mises stress (SI units, Pa) on the deformed shape (×100).

3 Square plate under sudden pressure (FEM workbench, shells)

Model and mesh. Plate 200 × 200 mm, thickness 2 mm, all edges clamped, $q = 0.01$ MPa applied at $t = 0$ and held (*Constant* load curve), no damping. $20 \times 20 = 400$ four-node shells (441 nodes), 10 mm squares, Belytschko–Tsay (`ELFORM 2`); span/thickness 100, so a thin plate.

Numerics. Time step 1.54 μs, 6 502 cycles, 0.8 s wall time, mass $6.32 \cdot 10^{-4}$ t as expected, energy error ≈ 0 %.

References. Static centre deflection of a clamped square plate $w = 0.00126\,qa^4/D = 0.131$ mm with $D = Et^3/(12(1-\nu^2))$. A load applied suddenly to an undamped system makes it oscillate between 0 and twice the static value. Fundamental frequency $f = 35.99/(2\pi a^2)\sqrt{D/(\rho t)} \approx 447$ Hz, i.e. a period of 2.24 ms.

Results. The centre deflection (Figure 2) oscillates between nearly 0 (0.002–0.011 mm) and 0.259–0.273 mm (twice the static value is 0.262). The period between peaks is 2.2 ms, and the average over the run is 0.134 mm (static theory 0.131, +2 %). All three agree with the references within a few per cent.

Mesh assessment. Sufficient for the first mode and the centre deflection. The frames are written every 0.2 ms (about 11 per period), so the extremes are resolved only to about ±3 %; write more frames to measure peaks more precisely. Real plates are damped, so the undamped oscillation persists longer than it would in a test.

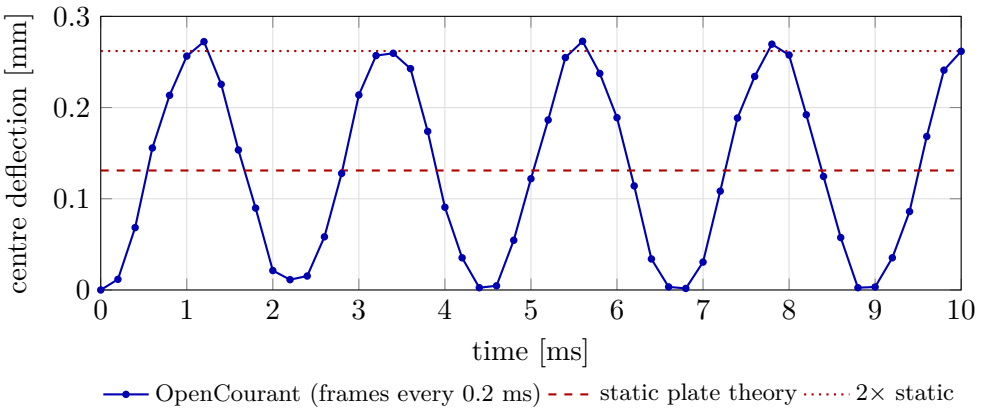


Figure 2: Plate centre deflection after the sudden pressure load.

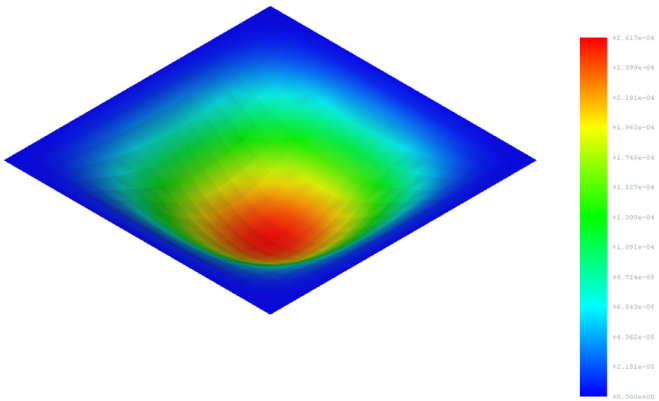


Figure 3: Plate at 10 ms, displacement magnitude (m), deformation ×200.

4 Taylor impact (keyword deck)

Model and mesh. Steel bar $10 \times 10 \times 40$ mm hitting a rigid wall (`*RIGIDWALL_PLANAR`) at $v = 200$ m/s, bilinear plasticity (`*MAT_PLASTIC_KINEMATIC`: yield 400 MPa, tangent modulus 1000 MPa), end time 60 μs. $4 \times 4 \times 16 = 256$ hexahedra (425 nodes), cubes of 2.5 mm, `ELFORM 1`.

Numerics. The time step drops from 0.158 to 0.152 μs as the elements at the impact end are compressed; 353 cycles, 0.1 s wall time; mass $3.14 \cdot 10^{-5}$ t as expected. Initial kinetic energy $\frac{1}{2}mv^2 = 6.28 \cdot 10^5$ N mm. At 52 μs the internal energy is $5.84 \cdot 10^5$ and the kinetic energy $2.2 \cdot 10^4$ (3.6 % of the initial value), so the bar has almost stopped. Internal plus kinetic energy is 3.4 % below the initial kinetic energy. A plausible explanation, not checked: the rigid wall stops the nodes of the impact face instantly; they carry about 1/32 (3.1 %) of the mass, and their kinetic energy is removed.

Results. The bar shortens from 40.0 to 32.2 mm ($L_1/L_0 = 0.805$). The impact end widens from 10.0 to 14.6 mm, and the equivalent plastic strain reaches 0.81 at the impact face. A rough consistency check is Taylor's (1948) formula $\sigma_y \approx \rho v^2 (L_0 - X)/(2(L_0 - L_1)\ln(L_0/X))$. With the undeformed length $X = 7.5$ mm taken from the simulation (the top part with plastic strain below 0.002) it gives $\sigma_y \approx 390$ MPa, close to the 400 MPa in the deck. Taylor's formula assumes rigid–perfectly plastic behaviour, so this checks plausibility only.

Mesh assessment. Four elements across are coarse for a Taylor test: the mushroom and the strain gradient at the impact face are spread over one or two elements. In the last frame the

largest element aspect ratio is 3.7 (median 1.5), and impact-face elements are compressed from 2.5 to 1.1 mm edge length. No element inverted, and the run terminated normally. For quantitative work refine to 8–12 elements across and check the hourglass energy of the one-point elements, which was not evaluated here.

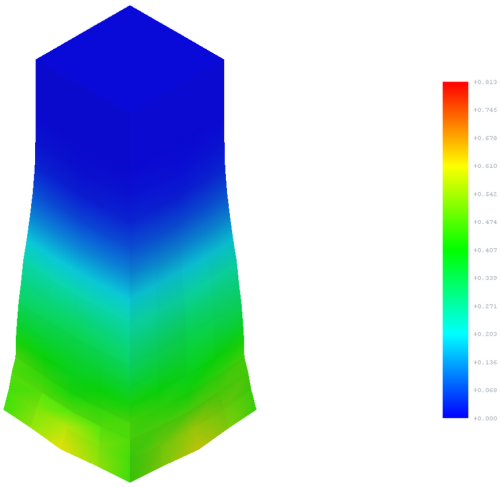


Figure 4: Taylor bar at 60 μs, equivalent plastic strain on the deformed shape (true scale).

5 Side study: tetrahedral meshes of the cantilever

The same cantilever was meshed with Gmsh and solved with OpenCourant and with CalculiX on the same mesh settings (Table 1). Linear tetrahedra lock in bending and are too stiff in both codes, but OpenCourant and CalculiX agree. With 10-node tetrahedra CalculiX matches beam theory while OpenCourant deflects about 35 % more, and its starter reports exactly 0.70 of the physical mass for every tetra10 mesh tried. The deck's tetra10 node order was checked: every other midside order is rejected by the starter. This points to the solver's 10-node element (or its LS-DYNA input mapping) rather than to the translation, but the cause has not been found.

Mesh (Gmsh)	Element	OpenCourant [mm]	CalculiX [mm]	Difference	Note
$h = 4$ mm	tetra4	0.1329	0.1307	+1.7 %	both locked
$h = 2$ mm	tetra4	0.1827	0.1679	+8.8 %	
$h = 4$ mm	tetra10	0.2592	0.1908	+36 %	no mass scaling, mass 0.70
$h = 2.5$ mm	tetra10	0.3072	0.1910	+61 %	mass scaling: total mass ×6

Table 1: Tip deflection of the cantilever with tetrahedra; beam theory 0.192 mm. The last row is influenced by the large added mass as well.

6 Summary

7 Limitations of this analysis

- Generated by an AI model and not verified by a human; listing values may have been misread and interpretations may be wrong.
- One run per case, coarse example meshes, no mesh or time-step convergence study (apart from the tetrahedra side study).

Case	Quantity	Simulation vs. reference	Rough assessment
Cantilever, hexa8	tip deflection	0.187 vs. 0.192 mm (−2.7 %)	good; stresses need a finer mesh
Plate, shells	period	2.2 vs. 2.24 ms	good
	peak / mean deflection	0.273 / 0.134 vs. 0.262 / 0.131 mm	good (frame spacing limits peaks)
Taylor bar	yield from Taylor's formula	390 vs. 400 MPa	plausible; mesh coarse
Cantilever, tetra10	tip deflection	0.259 vs. 0.191 mm (CalculiX)	not usable; solver issue open

- The references are simple analytical formulas (beam, plate, Taylor), each with its own assumptions; they check plausibility, not correctness.
- Hourglass energy, contact and wall forces and strain-rate effects were not evaluated.
- Results were read from the last frame or from frames every 0.2 ms and averaged to nodes by the workbench, which smooths peaks.

8 No warranty

No warranty

THIS DOCUMENT, THE EXAMPLES AND THE SOFTWARE ARE PROVIDED "AS IS", WITHOUT WARRANTY OF ANY KIND, EXPRESS OR IMPLIED, INCLUDING BUT NOT LIMITED TO THE WARRANTIES OF MERCHANTABILITY, FITNESS FOR A PARTICULAR PURPOSE, ACCURACY AND NON-INFRINGEMENT. THE ENTIRE RISK AS TO THE QUALITY, ACCURACY AND USE OF THE RESULTS IS WITH YOU. IN NO EVENT SHALL THE AUTHORS, CONTRIBUTORS OR COPYRIGHT HOLDERS BE LIABLE FOR ANY CLAIM, DAMAGES OR OTHER LIABILITY ARISING FROM, OUT OF OR IN CONNECTION WITH THIS DOCUMENT, THE SOFTWARE OR THEIR USE. Simulation results must be verified by a qualified engineer before they are used for any purpose. The workbench is licensed under the GNU LGPL 2.1 or later, which contains the same disclaimer; OpenCourant/OpenRadioss is licensed separately under the AGPL 3.0.